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IN-VITRO EVALUATION OF EFFLUX PUMP INHIBITORY ACTIVITY OF CHENOPODIUM ALBUM AGAINST ANTIBIOTIC-RESISTANT BACTERIA

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ABSTRACT:

Background: *Antimicrobial resistance (AMR) is a global healthcare challenge, partly driven by bacterial efflux pumps that expel antibiotics. This study evaluated Chenopodium album leaf extract as a potential efflux pump inhibitor (EPI) against multidrug-resistant (MDR) Escherichia coli.*

Methods: *E. coli isolates were assessed for antibiotic resistance according to CLSI guidelines. Leaves were extracted using ethanol and screened for phytochemicals. Antimicrobial activity and minimum inhibitory concentration (MIC) were determined using microbroth dilution, while an ethidium bromide (EtBr) accumulation assay assessed efflux inhibition.*

Results: *The extract contained alkaloids, flavonoids, and saponins. It modified antibiotic resistance, reducing ciprofloxacin MIC from 8 to 2 µg/mL, indicating a fourfold increase in susceptibility. Intracellular fluorescence in the EtBr assay suggested efflux inhibition and reduced drug extrusion.*

Conclusion: *Chenopodium album extract showed potential as a natural EPI and enhanced ciprofloxacin activity against MDR E. coli. These findings support further investigation of plant-derived compounds as potential adjuncts to conventional antibiotics.*

Keywords: *Antibiotic Resistance, Chenopodium album, Efflux Pump, Efflux Pump Inhibitors, Multidrug- E. Coli*

1. Introduction

The cornerstone Antibiotics have been modern medicine, saving the millions of since their discover lives. They allow to safe surgical procedures as well as cancer chemotherapy, and the life-threatening infections treatment. Yet, the growing problem of antimicrobial resistance (AMR) is the threatening these achievements. AMR occurs when the microorganisms adapt and to survive drugs exposure once killed them for the making harder infections to treat.

The declaration by World Health Organization (WHO) has AMR top ten global health threats. In 2019, directly responsible resistant infections for 1.27 million deaths, in worldwide with nearly 5 million deaths associated (Who,2023).This crisis is not limited to hospitals pathogens resistant in communities spreading, and in the environment.

Accelerate resistance Several drivers. in humans Misuse of antibiotics for viral infections such as prescribing them or incomplete treatment courses which creates selective to the pressure for resistant strains. In agriculture livestock antibiotics are the widely used for the growth promotion, which resistance genes contributes entering the food chain (Van Boeckel,2015). Poor infection control practices in the hospitals and also in inadequate sanitation in communities for the further situation worsen (Laxminarayan,2020).

The consequences are severe given below

a) untreatable common infections
b) carries high risks routine surgeries
c) need for stronger medical costs will rise
d) more expensive drugs.
e) the world may face a post-antibiotic without urgent intervention
f) could prove fatal where even minor illnesses (O'Neill, J,2016).

The misuse of antibiotics one of the most significant contributors in humans. demand antibiotics Patients often for viral infections such as influenza as well as common cold, where these drugs are ineffective. (Ventola, C. L.,2015).

the extensive another major driver is use of antibiotics in the agriculture. Livestock are frequently given antibiotics and also to promote growth and prevent disease in the crowded farming conditions. Into the food chain practice introduces resistant bacteria, which can to humans spread through direct contact or the consumption (Van Boeckel,2017).

Environmental factors also affected. hospital effluents, pharmaceutical waste, and improper disposal of the antibiotics contaminate water and the soil, creating reservoirs of resistance genes. These genes can be transfer between the bacterial species, accelerating the spread of the AMR (Kummerer, K.,2009).

Poor infection control practices in the healthcare settings such as inadequate hand hygiene as well as lack of isolation protocols, and broad-spectrum antibiotics overuse the transmission of resistant pathogens was facilitating. This is dangerous particularly in intensive care units, where patients vulnerable are exposed to multidrug-resistant organisms (Prestinaci, F.,2015).

Efflux pumps also to complicate treatment in the nosocomial infections. On efflux systems Pathogens such as *Acinetobacter baumannii* and *Klebsiella pneumoniae* rely heavily, which explains resistant frequently to multiple antibiotics in the intensive care units (Coyne, S.,2011). This leads to the prolonged hospital stays and increased mortality.

Major gap is development the limited of new antibiotics. Pharmaceutical companies have investment reduced in antibiotic discovery due to high costs and financial returns low. As a result, for novel drugs pipeline is shrinking, leaving clinicians with the fewer options by against resistant pathogens (Theuretzbacher,2020).

Another gap lies in the lack of the clinically approved efflux pump inhibitors (EPIs). many molecules have shown in the promise in laboratory studies, toxicity issues, specificity, and the pharmacokinetics have been prevented their widespread us in patients (Opperman, T. J.,2015). In this research highlights the interdisciplinary research with combining microbiology, pharmacology, as well as computational drug design.

2. Material and Method

2.1 Microbial Cultures and Culture Media

Clinical isolates of *Escherichia coli* were procured from Neuberg Diagnostic Centre, Raipur (Chhattisgarh). . For experimental use, the isolates were incubated in nutrient broth at 37 °C under aseptic conditions . For

microbial culture and maintenance, nutrient agar and nutrient broth were employed. These media were prepared according to standard microbiological protocols and sterilized by autoclave at 121 °C for 15 minutes before use.

2.2 Collection and Prepration of plant extract

Fresh leaves of *Chenopodium album* were collected on 14 January 2026 from agricultural fields in Palaud village, Raipur (Chhattisgarh). The plant material was authenticated by a qualified botanist. The collected leaves were thoroughly washed to ensure cleanliness. Then shade-dried at ambient temperature until completely free of moisture. The dried leaves were grind to achieve uniform particle size.

For extraction, the cold maceration method was employed. Precisely 25 g of powdered leaf material of *Chenopodium album* was weighed using a digital weighing balance, transferred into a sterile conical flask, and 250 mL of ethanol was added as the extraction solvent. The mixture was kept at ambient temperature for 72 hours with occasional shaking to facilitate maximum diffusion of bioactive compounds into the solvent. After the maceration period, the extract was filtered through Whatman No. 1 filter paper to remove insoluble plant residues. The filtrate was concentrated under reduced pressure at temperatures below 40 °C using a rotary evaporator to obtain a semi-solid crude extract. The dried extract was stored in an airtight container under refrigeration (4 °C) until further use in experimental assays Ahmed Ujan, J.,(2025)

2.3 Chemical Reagents

Ciprofloxacin powder as a representative antibiotic, while verapamil tablets served as a positive efflux pump inhibitor. Ethidium bromide powder as a fluorescent substrate. Ethanol as an extraction solvent. Sulfuric acid, sodium hydroxide, ferric chloride, and potassium iodide were used for qualitative phytochemical assays. Phosphate buffer, glucose, and normal saline .

2.4 Instruments

An autoclave was used for sterilization of media and glassware. A water bath was employed for controlled heating during assays. An incubator maintained at 37 °C was used for bacterial culture growth. A centrifuge was utilized for separation of supernatants and pellets. Micropipettes were used for accurate measurement of liquid volumes. A 96-well plate was employed for efflux pump inhibition assays

2.5 Preparation of Agar Plates

For antibiotic susceptibility testing, nutrient agar plates were prepared under sterile conditions. A measured quantity of 5.6 g nutrient agar powder was dissolved in 200 mL distilled water inside a 500 mL conical flask. The mixture was thoroughly mixed until the powder completely dissolved. The mouth of the flask was plugged with cotton and covered with paper to avoid contamination during sterilization.

The prepared medium was sterilized in an autoclave at 121 °C, 15 lbs pressure for 15 minutes (EUCAST,2024). After sterilization, the flask was carefully removed and allowed to cool slightly to about 45–50 °C. The molten agar was then aseptically poured into sterile Petri dishes inside the laminar air flow cabinet to maintain aseptic conditions (Flinn Scientific, 2023).

Once the agar solidified, the Petri dish lids were closed and sealed with parafilm to prevent contamination. The plates were stored at **4 °C in a refrigerator** until further use. This procedure ensured that the agar plates were sterile, contamination-free, and suitable for uniform bacterial growth .

2.6 Phytochemical Screening

The ethanol extract of *Chenopodium album* leaves was subjected to preliminary phytochemical screening to identify major classes of bioactive constituents. (Trease, G. E.,1989; Sofowora, A.,1993; Jigna, P.,2006). The extract was evaluated for the presence of alkaloids, saponins, tannins, flavonoids, terpenoids, and glycosides .

2.7 Antibiotic Susceptibility Test (AST)

multidrug-resistant *Escherichia coli* isolates were evaluated by using the method Kirby–Bauer disc diffusion under the sterile conditions. inside a laminar air flow cabinet All experimental procedures were performed to maintain aseptic environment to preventing contamination (CLSI,2025).

2.8 Minimum Inhibitory Concentration (MIC) Assay

Ciprofloxacin minimum inhibitory concentration (MIC) by plant extract *Chenopodium album* extract, with combinations determined the broth microdilution method using in sterile 96-well microplates. This selected

technique because it is widely accepted for the antimicrobial susceptibility testing and reproducible quantitative results are providing (CLSI,2023; Bench Chem,2025).

2.9 Ethidium Bromide (EtBr) Accumulation Assay

The EtBr accumulation assay was performed to evaluate the efflux pump inhibitory activity of *Chenopodium album* extract in comparison with verapamil (a known efflux pump inhibitor) and ciprofloxacin. This method is widely used to study bacterial efflux activity because EtBr is a fluorescent dye that enters cells and is expelled by efflux pumps (Paixao, L.,2019)

3. Result

3.1 Phytochemical Analysis Extract of the *Chenopodium album*

Ethanolic leaf extract of *Chenopodium album,* Preliminary phytochemical screening was carried out to identify the major bioactive secondary metabolites present. These phytoconstituents contribute significantly to the pharmacological properties and resistance-modifying activities particularly their antimicrobial. these identification compounds provide a scientific basis for the evaluating the potential role of the extract including efflux pump inhibition.

Table 3.1 - *Chenopodium album* Extract Detected of Phytochemical Constituents

S. No.	Phytochemical Constituent	Reagent/ Test Applied	Observation	Result
1	Alkaloids	Wager test	Reddish brown ppt	+
2	Saponins	Foam test	Foam occur	+
3	Tannins	Ferric chloride + extract	Blue-black or green coloration	+

4	Flavonoids	Alkaline reagent	Development of yellow coloration	+
5	Terpenoids	Salkouski test	Slight yellow color	+
6	Glycosides	Keller-Killani test	Producing slight brown ring, blue green upper layer	+

3.2 Antibiotic Susceptibility Test (AST)

The susceptibility profile antibiotic of the multidrug-resistant *Escherichia coli* isolate was determined the Kirby–Bauer disc diffusion method using in the accordance with the standard protocol. This performed assay to evaluate the response of the bacterial isolate to the commonly used in the antibiotics and to the confirm the presence of resistance.

Table 3.2 - Antibiotic Susceptibility Pattern

Antibiotic	**Potency Disc**	**Zone of Inhibition in mm measuring**	**Interpretation**
Ciprofloxacin	5 µg	22 mm	Sensitive
Amoxicillin	10 µg	12 mm	Resistant

3.3 Minimum Inhibitory Concentration (MIC) Assay

Ciprofloxacin minimum inhibitory concentration (MIC), *Chenopodium album* extract, and their determined combinations using the broth microdilution method. This performed assay was to evaluate the antibacterial activity of the test agents and to investigate the possible role of the plant extract in the modulating bacterial resistance to the mechanisms.

Table 3.3 - MIC Values of Test Samples

Sample	MIC (µg/mL)	Observation
Ciprofloxacin	8	Growth inhibited at moderate concentration
C. album extract	32	Inhibition observed at higher concentration
Ciprofloxacin + Extract	2	Inhibition at lower concentration
Ciprofloxacin + Verapamil	1	Maximum inhibition observed

3.4 Ethidium Bromide (EtBr) Accumulation Assay

The ethidium bromide (EtBr) accumulation assay was performed to evaluate the efflux pump inhibitory potential of *Chenopodium album* extract in multidrug-resistant *Escherichia coli*. This assay is based on the principle that EtBr is a fluorescent dye capable of entering bacterial cells and intercalating with DNA. However, in the presence of active efflux pumps, EtBr is rapidly expelled from the cell, resulting in reduced intracellular accumulation and weak fluorescence.

Table 3.4 - Fluorescence Observation

S. No.	Tube	Treatment	Fluorescence Intensity	Interpretation
1	Tube 1	Control	Low	Active efflux present
2	Tube 2	Verapamil	High	Efflux pump inhibited
3	Tube 3	*C.album* extract	Moderate to High	Efflux inhibition suspected

4	Tube 4	*Antibiotic*	Moderate	Antibiotic effect comparison

4. Discussion

This study looked at whether a leaf extract from Chenopodium album could help make antibiotics work better against E. Coli bacteria that're resistant to drugs. The results were good in two ways.

First when the leaf extract was used with the ciprofloxacin, the antibiotic only needed one fourth of its usual amount to stop the bacteria from growing. This means the extract is making the antibiotic stronger it is not working on its own.

Second a test using bromide showed that bacteria treated with the extract had a hard time removing the dye from their cells compared to bacteria that were not treated. Since bacteria use the method to remove antibiotics this suggests that the extract is blocking this method.

The Chenopodium album plant has things like flavonoids, alkaloids and saponins in it. We know these things can attach to the proteins that bacteria use to pump out things and they can interfere with how these proteins work. This is probably why the bacteria had a time pumping out the dye.

One thing to remember is that this study was only done in a lab. We still need to see how the extract works in a body. Can it get to the place where it is needed how long does it last and is it safe to use.

Overall Chenopodium album seems like it could be a way to help make antibiotics work better against bacteria that are resistant, to them. Chenopodium album could help restore the power of antibiotics that bacteria have learned to resist.

5. Coclusion

The present dissertation was undertaken to explore the potential of *Chenopodium album* as a natural source of efflux pump inhibitors to combat the growing menace of multidrug resistance in *Escherichia coli*. Based on the systematic in-vitro evaluation conducted throughout this study, several significant conclusions can be drawn. Firstly, the preliminary phytochemical screening of the ethanolic extract of *Chenopodium album* revealed a rich presence of secondary metabolites, including alkaloids, flavonoids, and saponins. These bioactive constituents are scientifically recognized for their ability to interact with bacterial cell membranes

and modulate transport proteins, providing a solid chemical foundation for the observed antimicrobial and resistance-modifying activities.

A pivotal finding of this research is the synergistic potential of the plant extract when used in combination with conventional antibiotics. The Minimum Inhibitory Concentration (MIC) assays clearly demonstrated that the presence of *Chenopodium album* extract significantly lowered the concentration of Ciprofloxacin required to inhibit the growth of MDR *E. coli*. This reduction in MIC indicates that the extract acts as a sensitizing agent, effectively reversing the resistance mechanism of the bacteria and restoring the potency of the antibiotic. Such synergy is crucial in a clinical context, as it allows for the use of lower antibiotic doses, thereby reducing the risk of side effects and further development of resistance.

Furthermore, the mechanistic aspect of this study was validated through the Ethidium Bromide (EtBr) accumulation assay. The marked increase in fluorescence intensity within the bacterial cells treated with the extract provided direct evidence of efflux pump inhibition. Since EtBr is a known substrate for bacterial efflux systems, its intracellular retention confirms that the phytochemicals in *Chenopodium album* successfully blocked the transport proteins responsible for expelling toxic substances. This action mirrors the effect of standard inhibitors like Reserpine and Verapamil. Therefore, it is concluded that *Chenopodium album* possesses potent efflux pump inhibitory (EPI) properties, making it a promising natural candidate for the development of antibiotic adjuvants.

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Conflict of Interest

The authors declare that there are no conflict of interest related to this work.

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